

States of Matter

The Three Common States

Solids have a fixed shape and volume because their particles are tightly packed and only vibrate in place.

Liquids have a fixed volume but take the shape of their container, since their particles can move past one another while remaining close together.

Gases have neither a fixed shape nor a fixed volume, as their particles move freely and are spaced far apart.

Phase Changes

Melting is the transition from solid to liquid, while freezing is the reverse. Boiling (or vaporization) is the transition from liquid to gas, while condensation is the reverse.

Sublimation is the direct transition from solid to gas without passing through the liquid state, and deposition is the reverse.

Kinetic Molecular Theory

Kinetic molecular theory explains that temperature is a measure of the average kinetic energy of particles, and that increasing temperature increases particle motion, which can drive phase changes.